

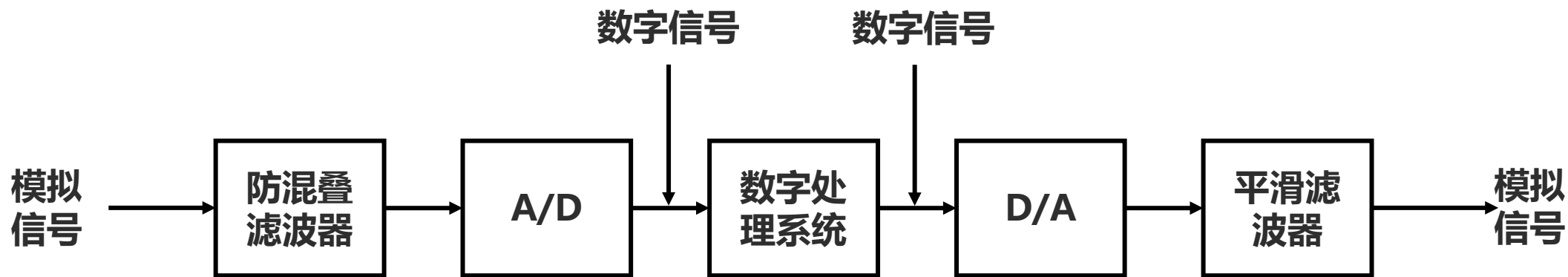
06 信号的采样

模拟信号和数字信号的互相转换



信号处理与分析的典型过程

- 数字信号的处理 (A:Analog, D:Digital)



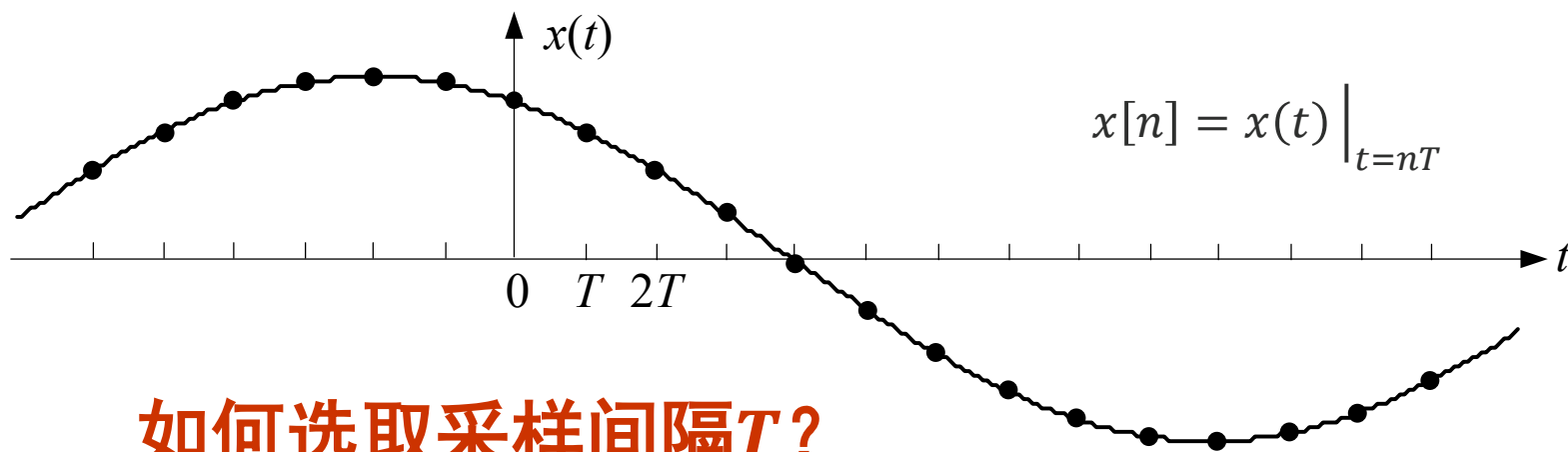
- 如何将模拟信号转化为数字信号?
- 如何将数字信号转化为模拟信号?

如何进行信号采样

- 图片信号



- 时序信号



为什么进行信号采样

- (1) 信号稳定性好: 数据用二进制表示, 受外界影响小。
- (2) 信号可靠性高: 存储无损耗, 传输抗干扰。
- (3) 信号处理简便: 信号压缩, 信号编码, 信号加密等
- (4) 系统精度高: 可通过增加字长提高系统的精度。
- (5) 系统灵活性强: 改变系统的系数使系统完成不同功能。



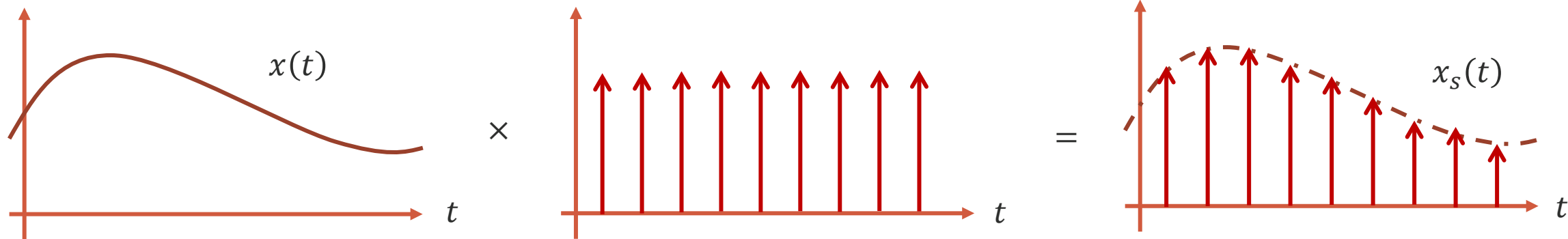
用数字方式处理模拟信号

信号的时域采样

- 信号 $x(t)$ 使用脉冲序列

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

采样得到 $x_s(t)$



$$x_s(t) = x(t) \cdot \delta_{T_s}(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s)$$

时域采样的频域分析

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

■ 已知

$$x_s(t) = x(t) \cdot \delta_{T_s}(t)$$

■ 设

$$\mathcal{F}[x(t)] = X(j\omega)$$

■ 则

$$\mathcal{F}[\delta_{T_s}(t)] = ?$$

周期信号的傅里叶变换

时域采样的频域分析

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

$$\mathcal{F}[x(t)] = \mathcal{F}\left[\sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}\right] = 2\pi \sum_{n=-\infty}^{\infty} X_n \delta(\omega - n\omega_0)$$

- 已知

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

求解 $\mathcal{F}[\delta_{T_s}(t)]$

$$\omega_s = \frac{2\pi}{T_s}$$

- 针对周期信号首先计算**傅里叶级数**

$$X_n = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta_{T_s}(t) e^{-jn\omega_s t} dt = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) e^{-jn\omega_s t} dt = \frac{1}{T_s}$$

傅里叶变换为

$$\mathcal{F}[\delta_{T_s}(t)] = 2\pi \sum_{n=-\infty}^{\infty} X_n \delta(\omega - n\omega_s) = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

时域采样的频域分析

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

- 已知 $x_s(t) = x(t) \cdot \delta_{T_s}(t)$, 设 $\mathcal{F}[x(t)] = X(j\omega)$, 且

$$\mathcal{F}[\delta_{T_s}(t)] = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

周期信号的傅里叶变换

- 由频域卷积特性

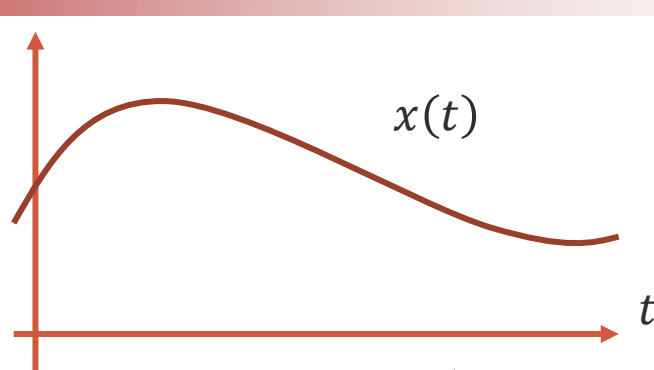
$$\omega_s = \frac{2\pi}{T_s}$$

$$\begin{aligned} \mathcal{F}[x_s(t)] &= \frac{1}{2\pi} \left[X(j\omega) * \mathcal{F}[\delta_{T_s}(t)] \right] = \frac{1}{2\pi} \left[X(j\omega) * \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s) \right] \\ &= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(j(\omega - n\omega_s)) \end{aligned}$$

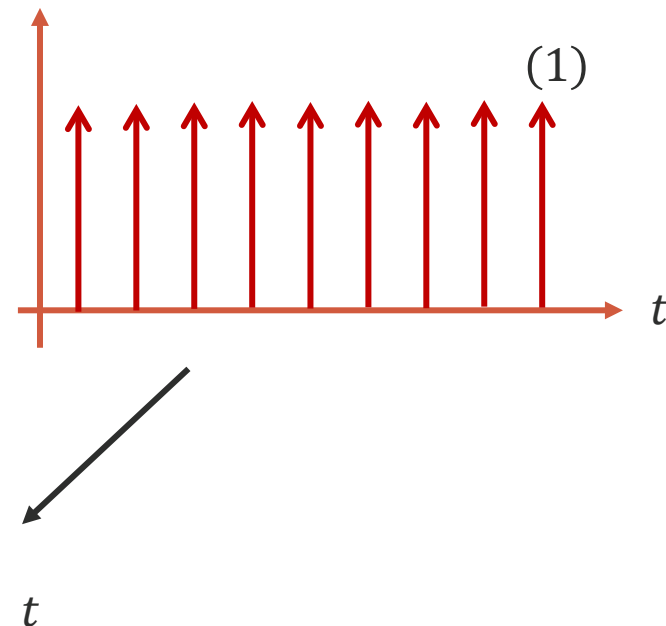
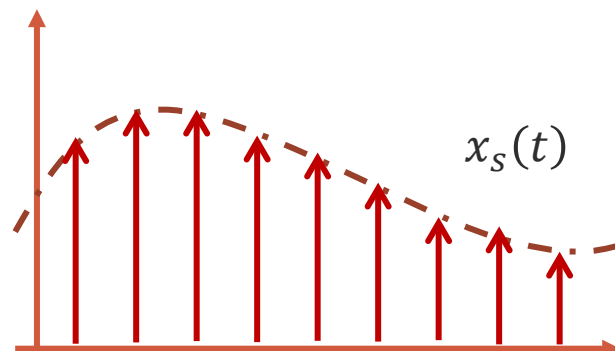
- 时域**对信号做**离散化**, **频域**表现为原始时域信号频谱 $X(j\omega)$ 的**周期延拓 (重复)**, 时域的离散化导致了频域的周期性

时域采样的频域分析

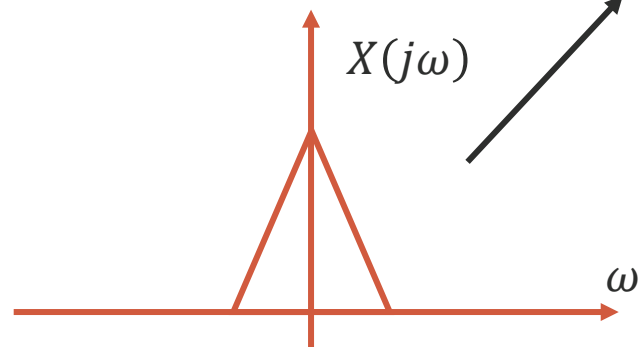
■ 时域



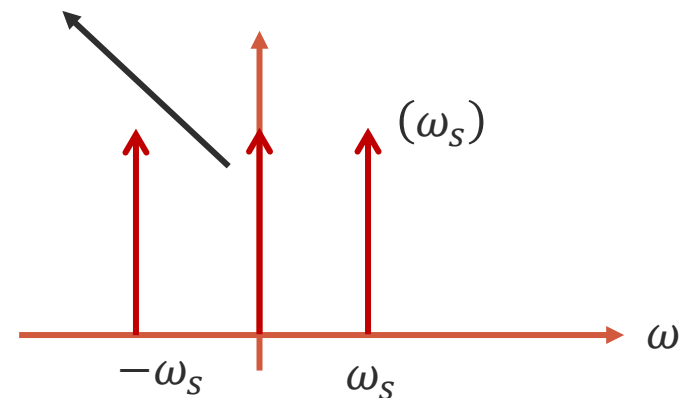
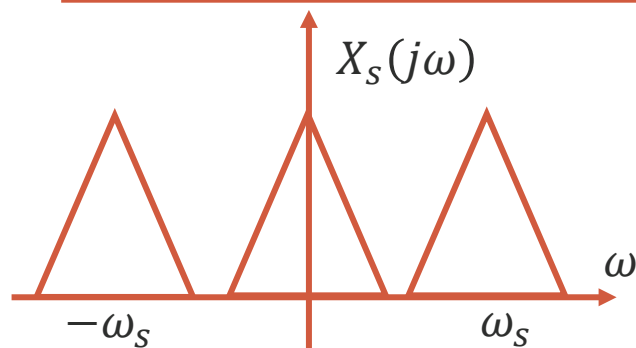
相乘



■ 频域



卷积



信号的时域采样

采样信号为周期矩形信号 $p(t)$ ，求采样后信号的频谱

- 根据 $x_s(t) = x(t) \cdot p(t)$ ，且 $p(t)$ 的傅里叶级数为

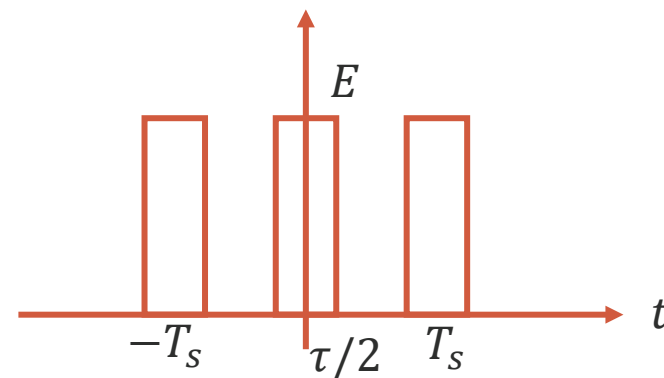
$$P_n = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} p(t) e^{-jn\omega_s t} dt = \frac{E\tau}{T_s} \text{Sa}\left(\frac{n\omega_s \tau}{2}\right)$$

因此

$$\mathcal{F}[p(t)] = 2\pi \sum_{n=-\infty}^{\infty} P_n \delta(\omega - n\omega_s)$$

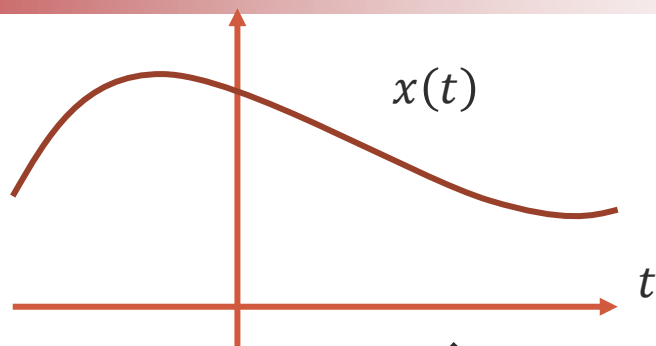
- 所以

$$\mathcal{F}[x_s(t)] = \sum_{n=-\infty}^{\infty} P_n X(j(\omega - n\omega_s))$$

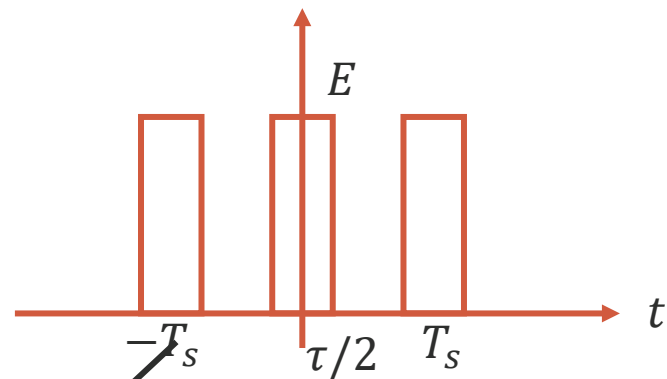
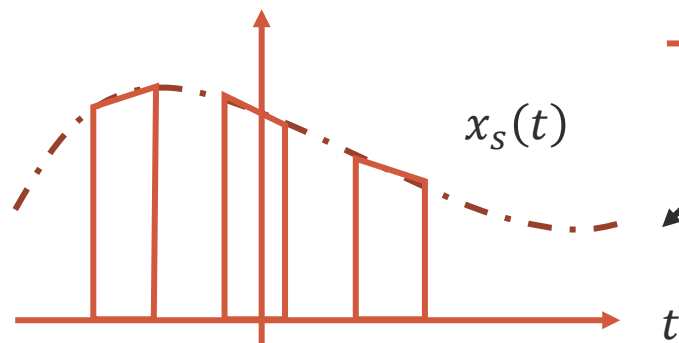


时域采样的频域分析

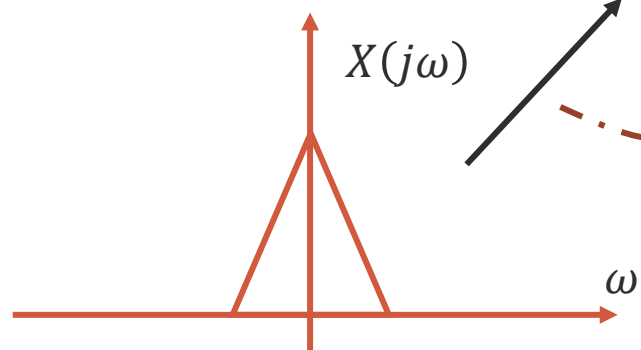
■ 时域



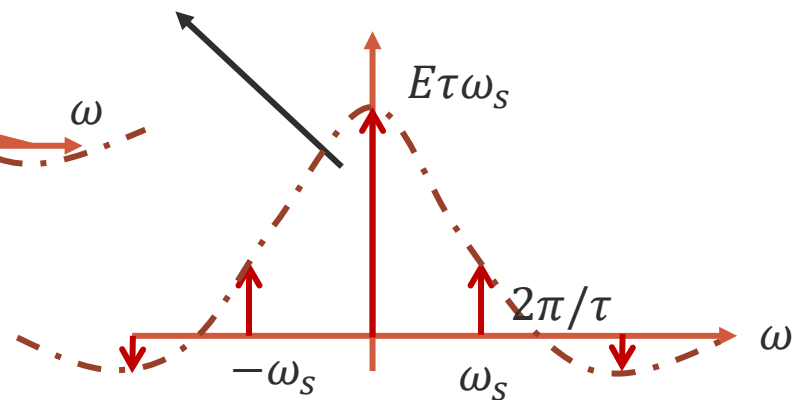
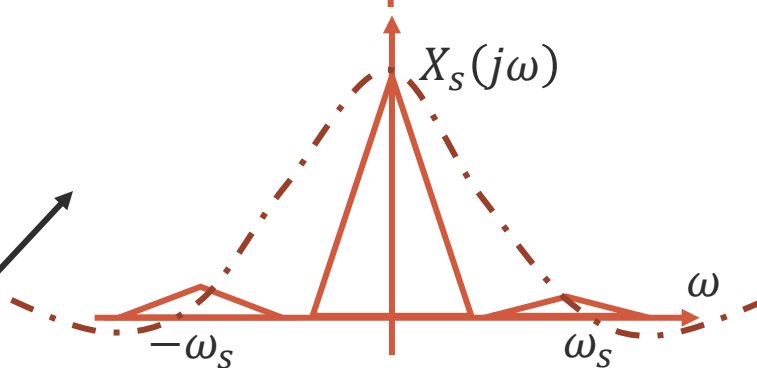
相乘



■ 频域



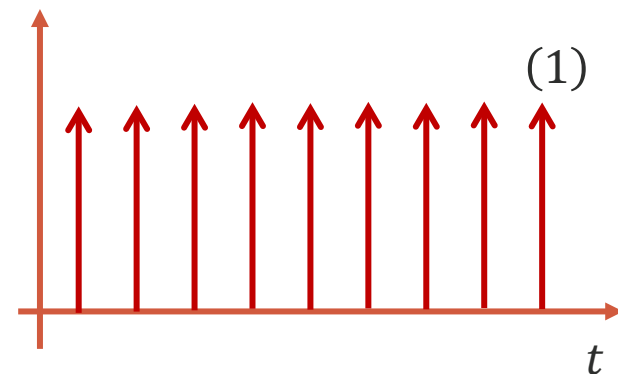
卷积



信号的时域采样

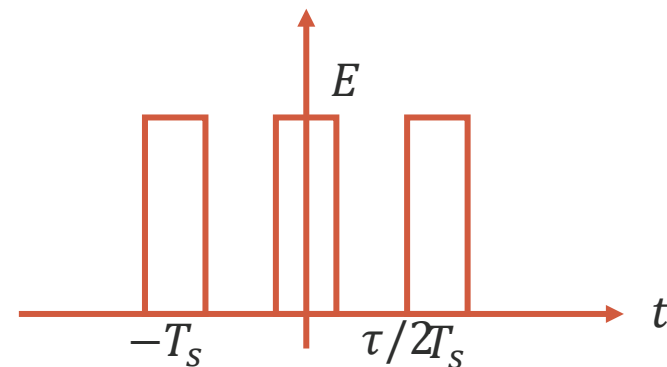
- 【理想采样】采样信号为周期冲激信号 $\delta_{T_s}(t)$ ，采样后信号的频谱

$$\mathcal{F}[x_s(t)] = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(j(\omega - n\omega_s))$$



- 采样信号为足够窄的周期矩形信号 $p(t)$ ，采样后信号的频谱

$$\mathcal{F}[x_s(t)] = \sum_{n=-\infty}^{\infty} P_n X(j(\omega - n\omega_s))$$



信号的频域采样

- 频域信号 $X(j\omega)$ 使用频域脉冲序列

$$\delta_{\omega_s}(\omega) = \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

采样得到 $X_s(j\omega)$, 因此

$$X_s(j\omega) = X(j\omega) \cdot \delta_{\omega_s}(\omega)$$

- 设 $\mathcal{F}^{-1}[X(j\omega)] = x(t)$, $\mathcal{F}^{-1}[X_s(j\omega)] = x_s(t)$, 由于

$$\mathcal{F}[\delta_{T_s}(t)] = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s) = \omega_s \delta_{\omega_s}(\omega)$$

$$\text{所以 } \mathcal{F}^{-1}[\delta_{\omega_s}(\omega)] = \frac{1}{\omega_s} \delta_{T_s}(t)$$

- 设

$$\mathcal{F}^{-1}[X(j\omega)] = x(t)$$

$$\mathcal{F}^{-1}[X_s(j\omega)] = x_s(t),$$

$$\mathcal{F}^{-1}[\delta_{\omega_s}(\omega)] = \frac{1}{\omega_s} \delta_{T_s}(t)$$

- 由卷积定理

$$x_s(t) = x(t) * \mathcal{F}^{-1}[\delta_{\omega_s}(\omega)]$$

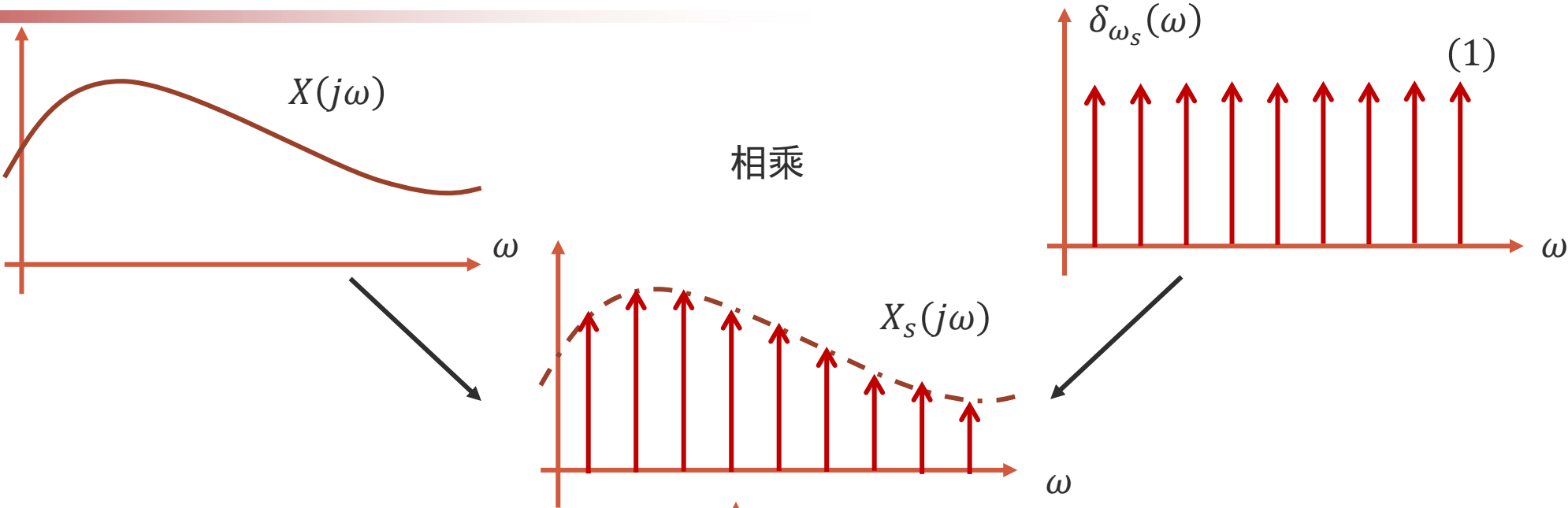
$$= x(t) * \left[\frac{1}{\omega_s} \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right]$$

$$= \frac{1}{\omega_s} \sum_{n=-\infty}^{\infty} x(t - nT_s)$$

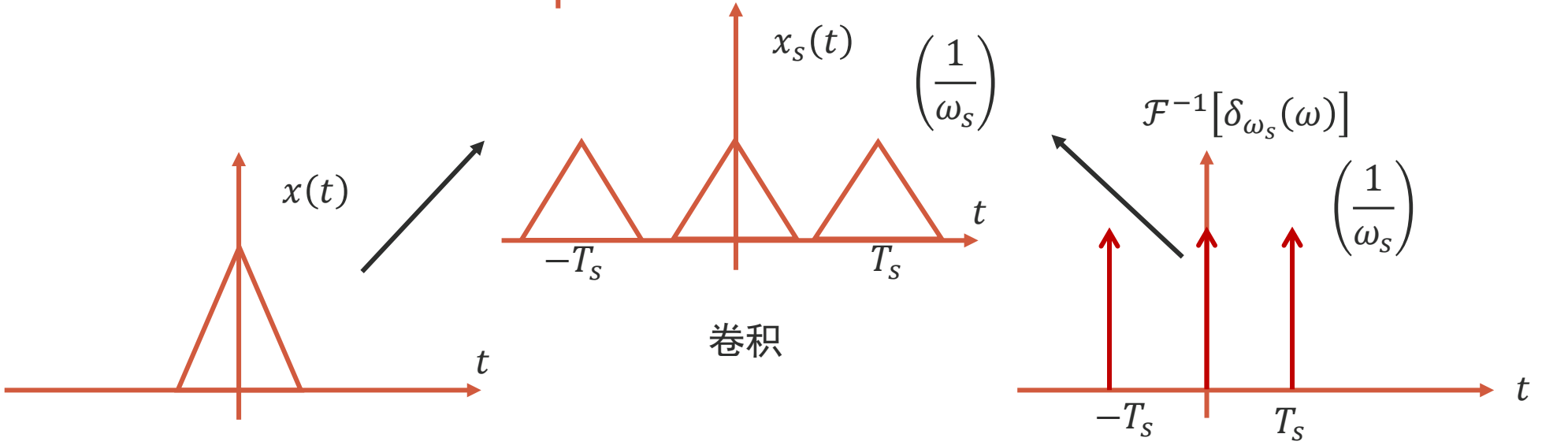
频域的离散化对应时域信号的周期延拓

信号的频域采样

■ 频域

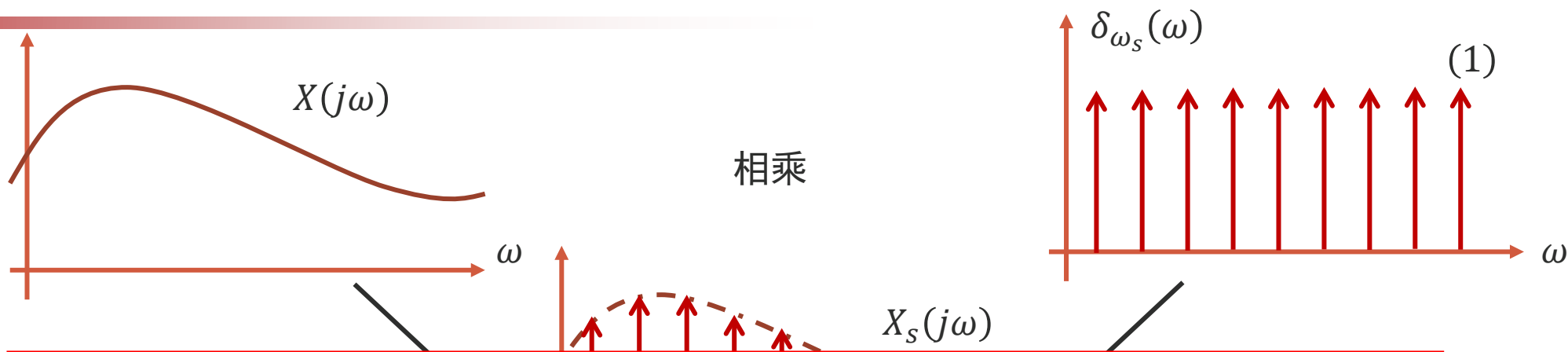


■ 时域



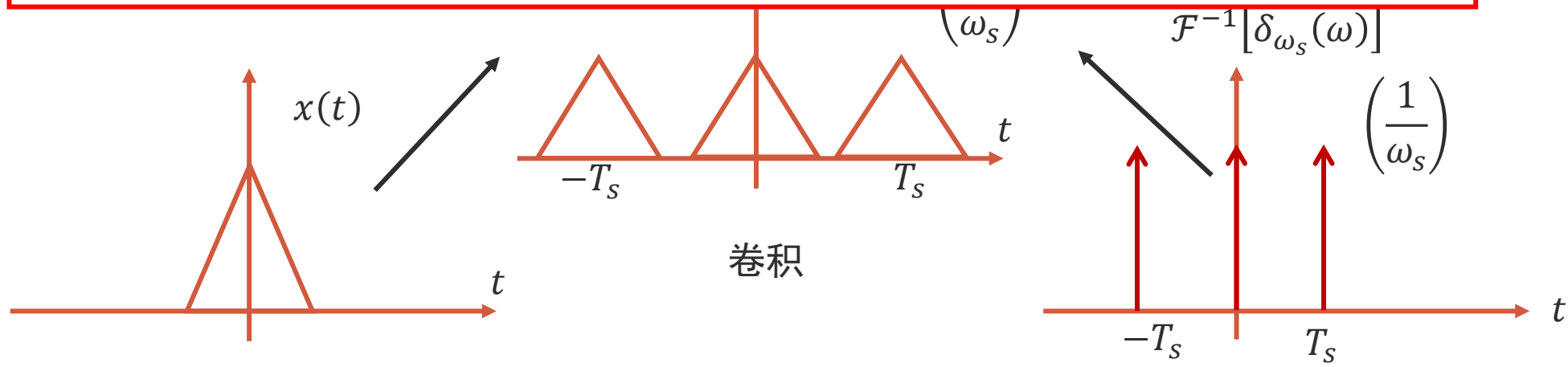
信号的频域采样

■ 频域



■ 时域

一个域的**离散化**和另一个域的**周期性**相对应，反之，一个域的**连续性**与另一个域的**非周期性**相对应



时域采样定理

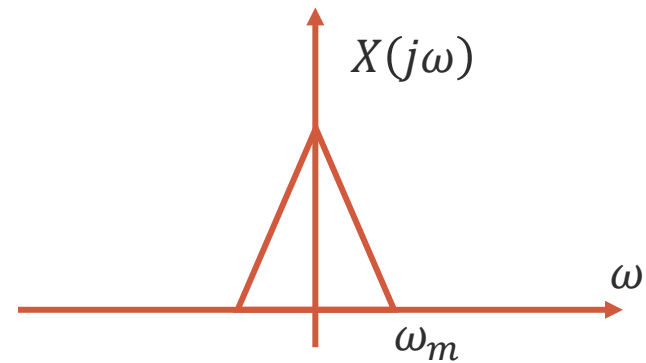
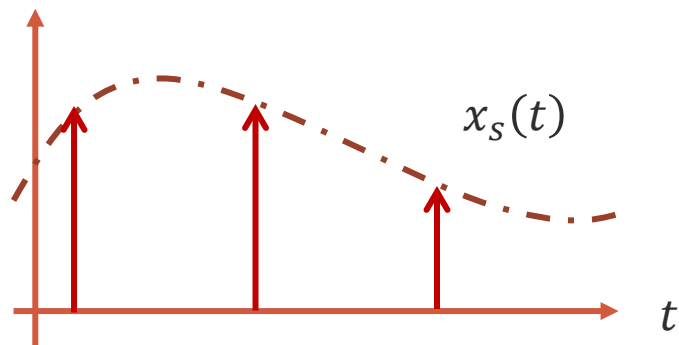
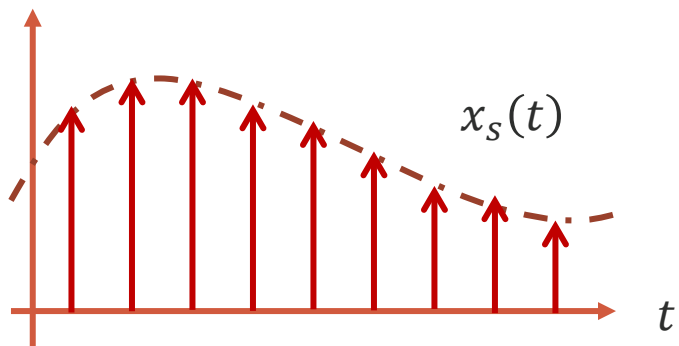
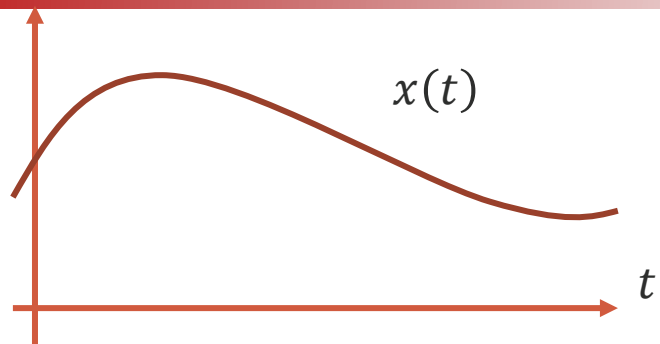
- 采样定理：若**连续**信号 $x(t)$ 是一个**频带受限**信号（若 $|\omega| > \omega_m$ 则 $X(j\omega) = 0$ ， $\omega_m = 2\pi f_m$ ）， $x(t)$ 的等间隔样本值 $x_s(t)$ ，用 $x_s(t)$ **唯一表示** $x(t)$ 的条件是

$$T_s < \frac{1}{2f_m}, \quad \text{即} \omega_s > 2\omega_m$$

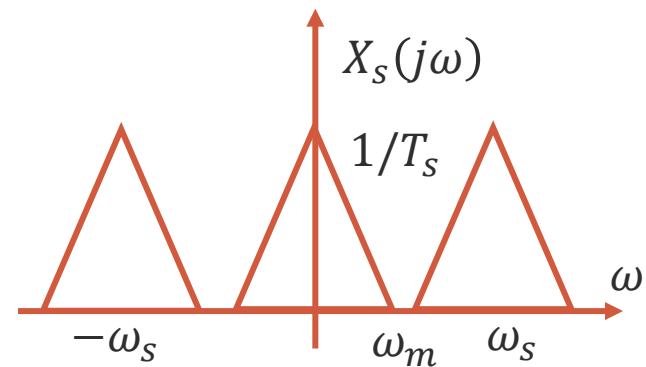
$f_s = 2f_m$ 为最小采样频率，称为Nyquist Rate.

- 条件
 - 等间隔采样，频带受限信号
 - 唯一恢复条件（采样频率）
 - 恢复方法（低通滤波器）

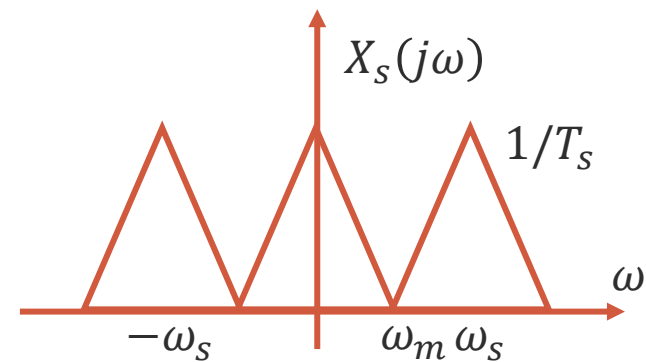
信号的时域采样



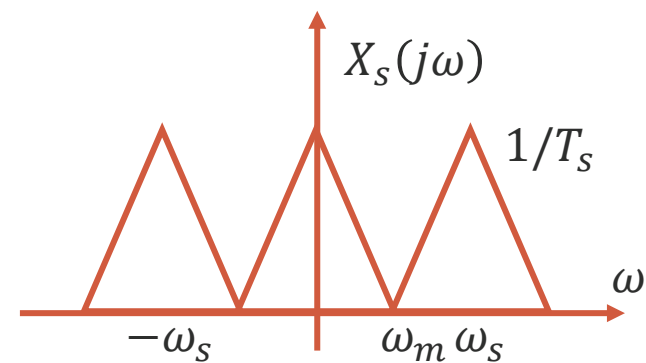
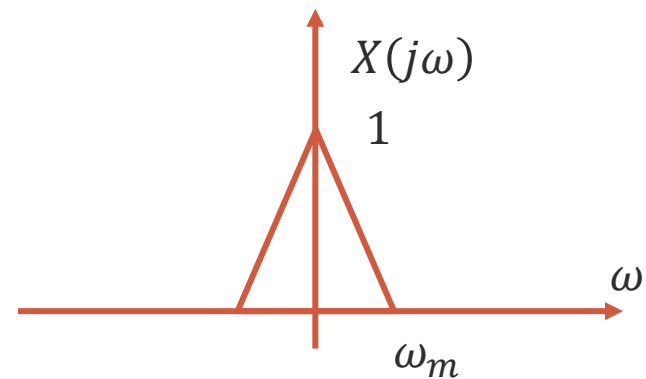
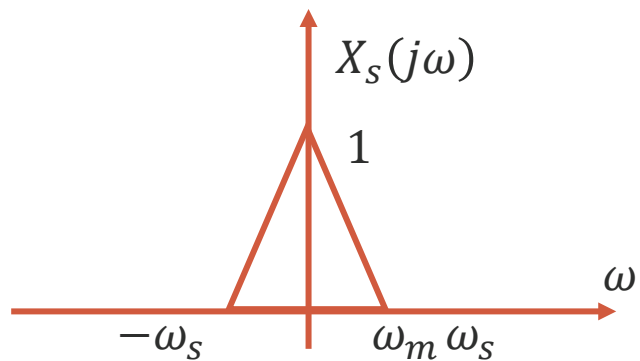
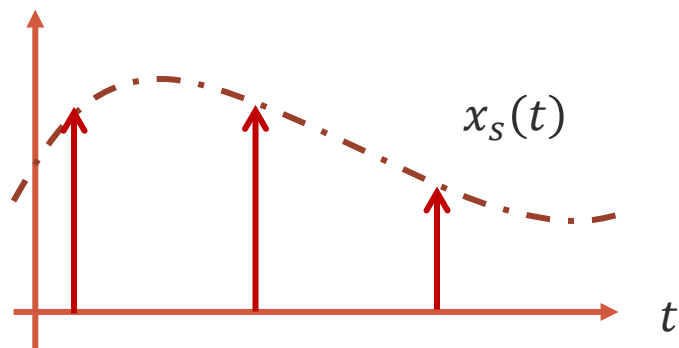
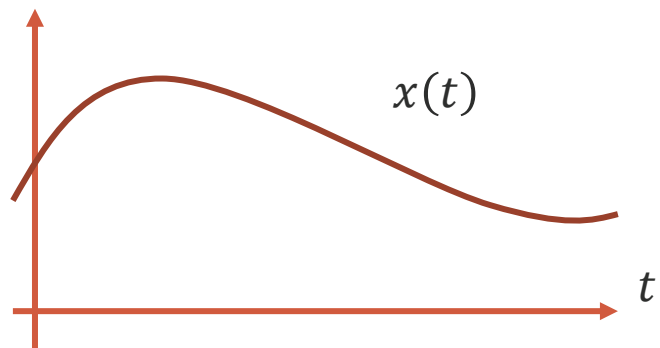
$$\omega_s > 2\omega_m$$



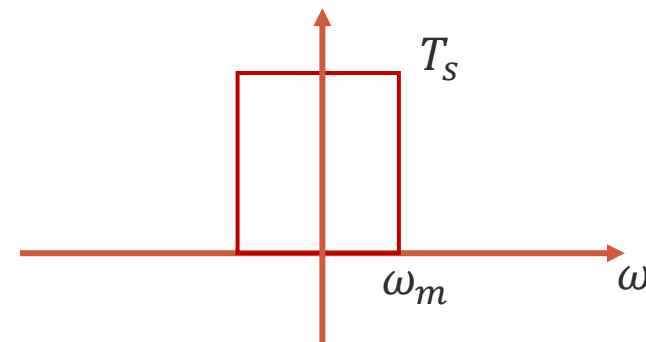
$$\omega_s = 2\omega_m$$



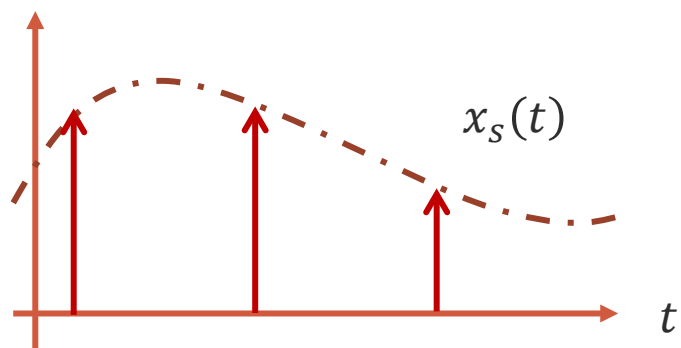
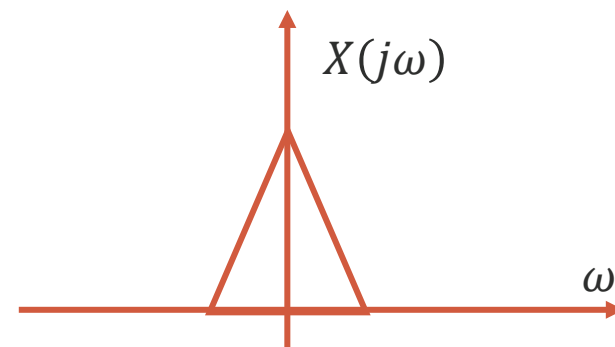
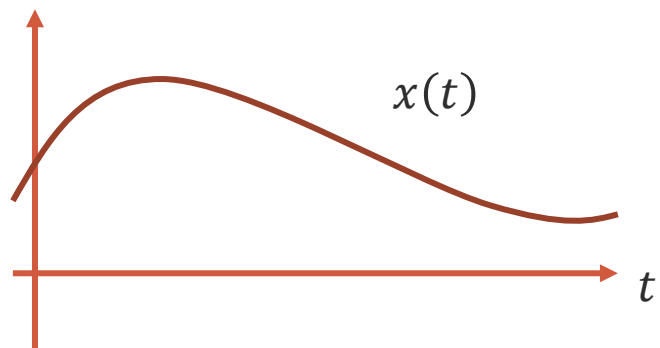
信号时域采样的恢复



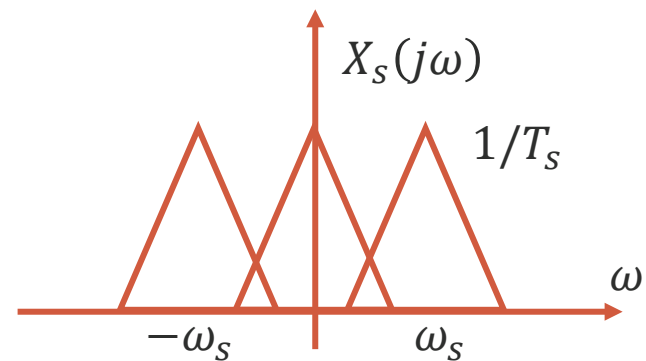
相乘
←



信号的时域采样



$$\omega_s < 2\omega_m$$



产生混叠 (aliasing)

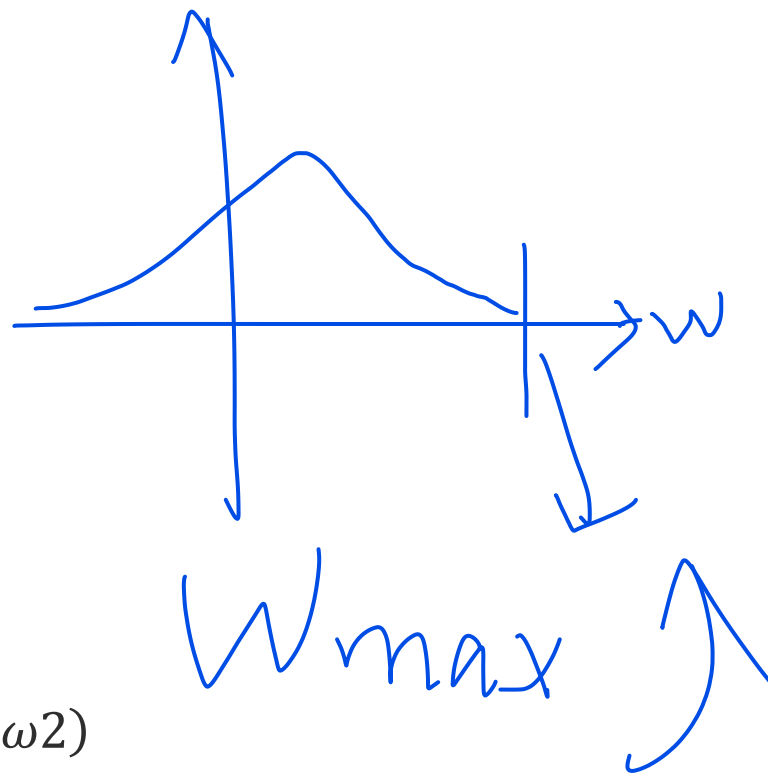
信号的时域采样

已知实信号 $x(t)$ 的最高频率为 f_m (Hz)，试计算对各信号 $x(2t)$, $x(t) * x(2t)$, $x(t) \cdot x(2t)$ 采样不重叠的最小采样频率。

- 对信号 $x(2t)$ 采样时，最小采样频率为 $4f_m$ (Hz)
- 对 $x(t) * x(2t)$ 采样时，最小采样频率为 $2f_m$ (Hz)
- 对 $x(t) \cdot x(2t)$ 采样时，最小采样频率为 $6f_m$ (Hz)

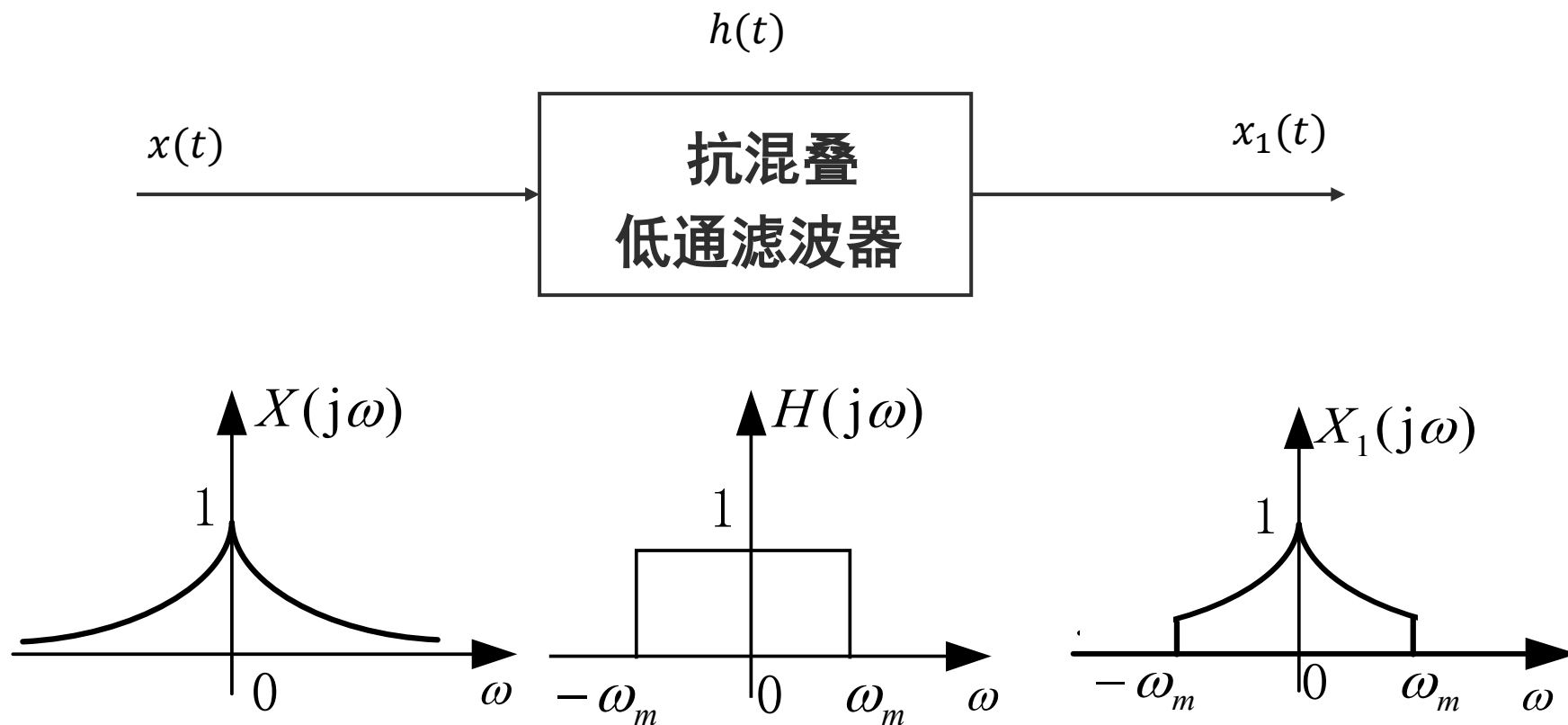
时域相乘相当于频域卷积，所以带宽为 $\omega_1 + \omega_2$

时域卷积相当于频域相乘，所以带宽为 $\min(\omega_1, \omega_2)$



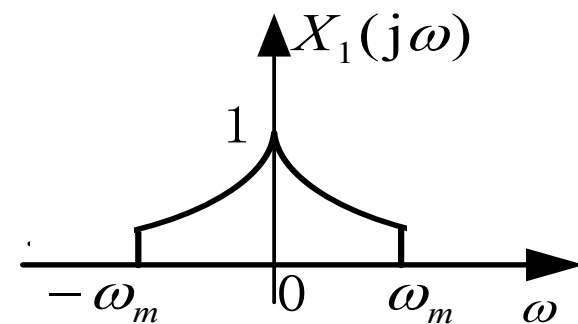
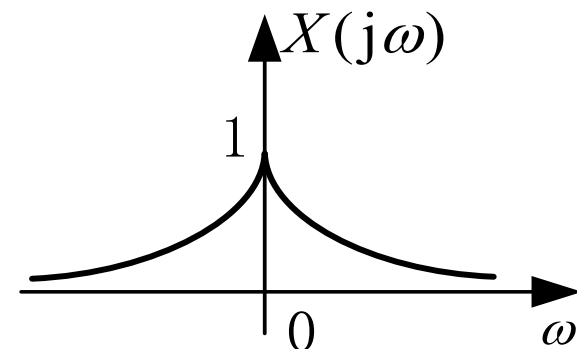
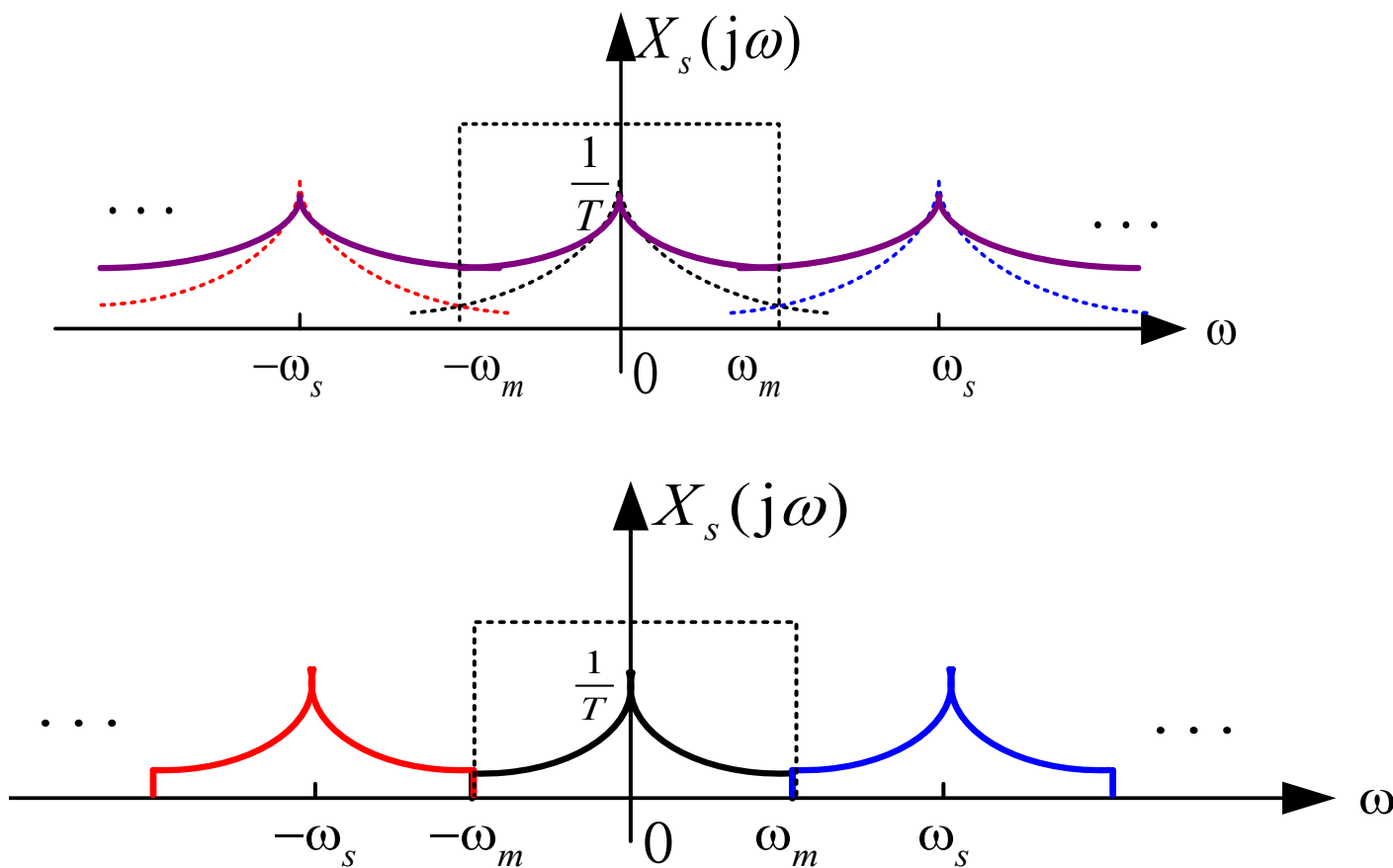
采样定理的实际应用

- 许多实际工程信号不满足带限条件



采样定理的实际应用

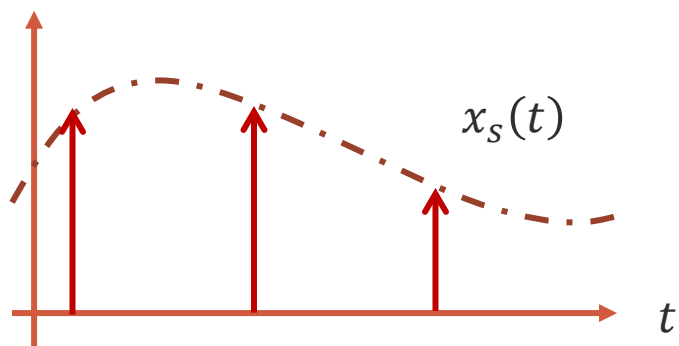
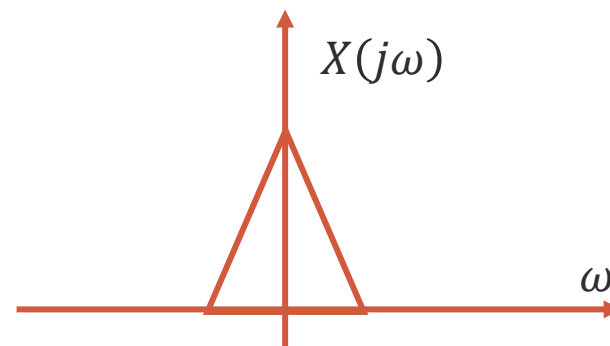
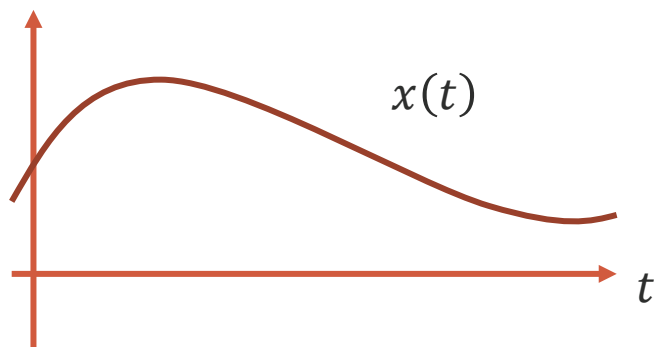
■ 混叠误差与截断误差比较



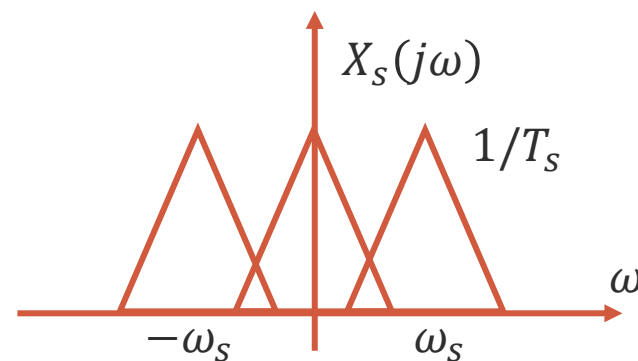
为避免频谱混叠，通常把采样频率设为 $\omega_s = (3 \sim 4)\omega_m$ 或更高

信号的时域采样 (欠采样)

- 不满足采样定理条件时，信号重构时可能会干扰原始频谱



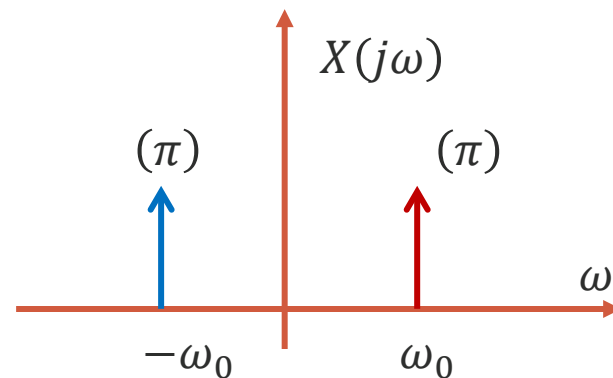
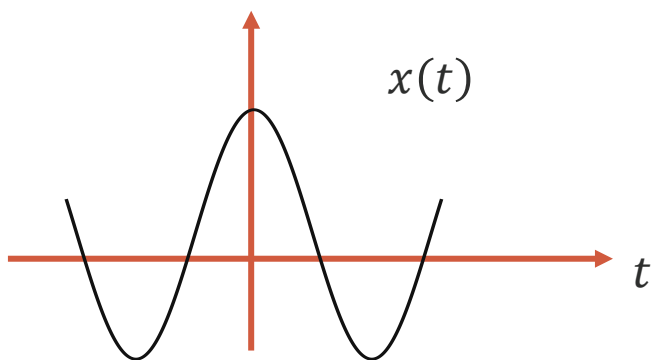
$$\omega_s < 2\omega_m$$



产生混叠 (aliasing)

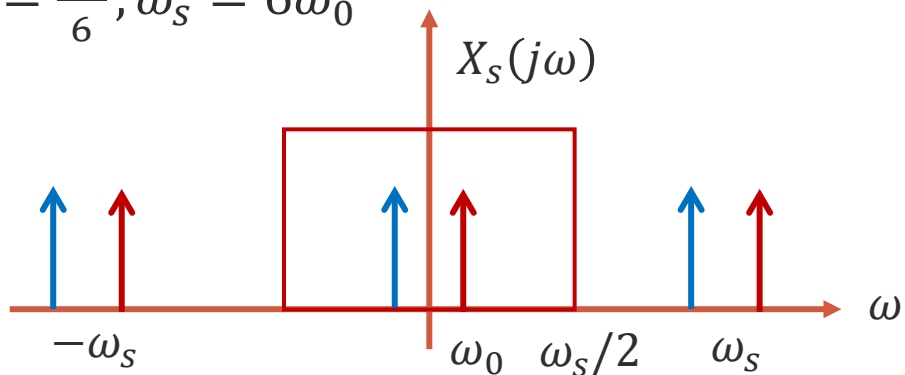
信号的时域采样 (欠采样)

- 设 $x(t) = \cos(\omega_0 t)$

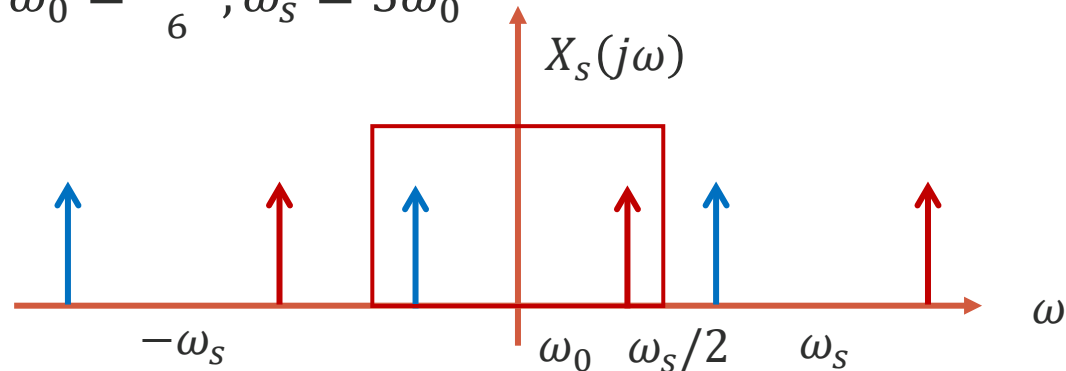


- 固定 ω_s** , 考察不同 ω_0 与 ω_s 的关系时的情况

$$\omega_0 = \frac{\omega_s}{6}, \omega_s = 6\omega_0$$

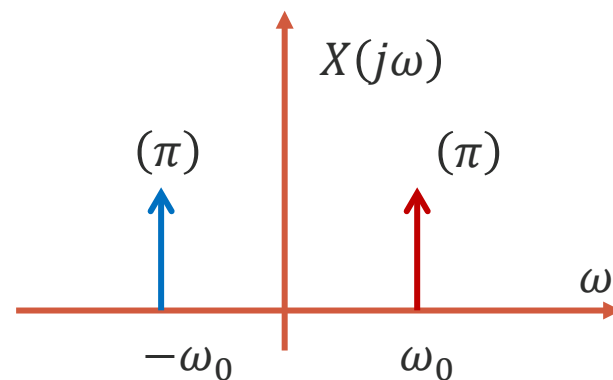
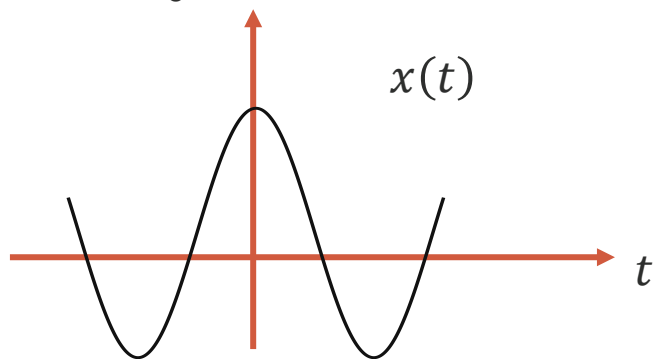


$$\omega_0 = \frac{2\omega_s}{6}, \omega_s = 3\omega_0$$



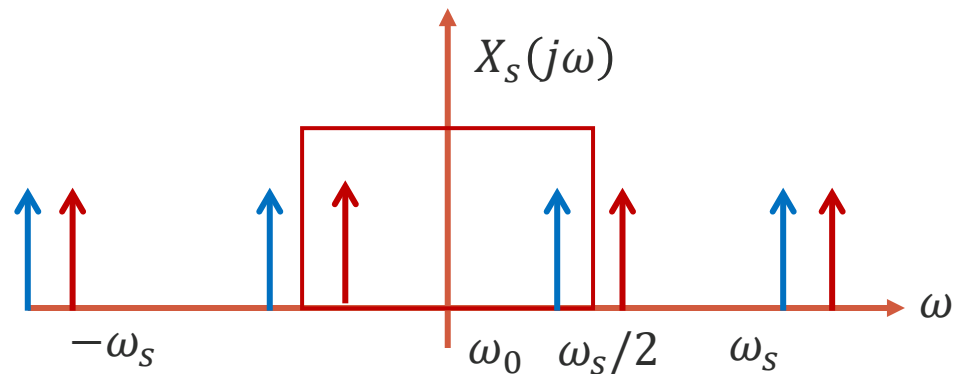
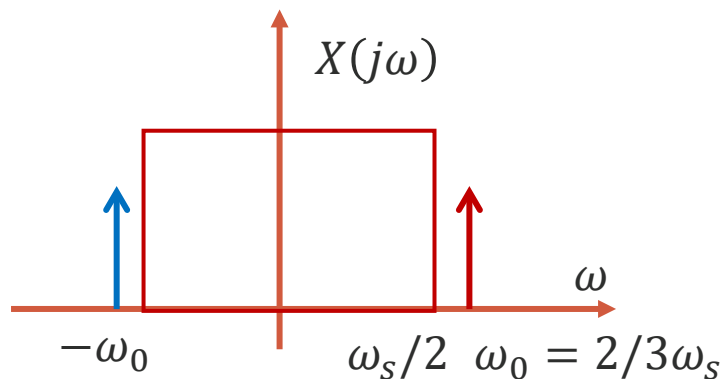
信号的时域采样 (欠采样)

- 设 $x(t) = \cos(\omega_0 t)$



- 固定 ω_s , 考察不同 ω_0 与 ω_s 的关系时的情况

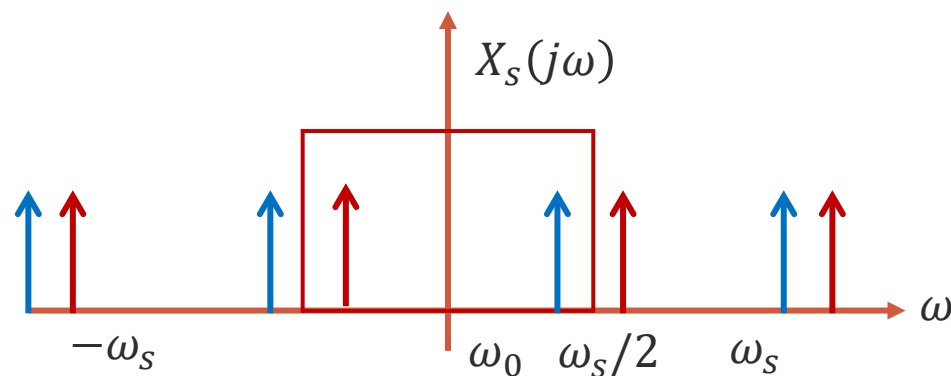
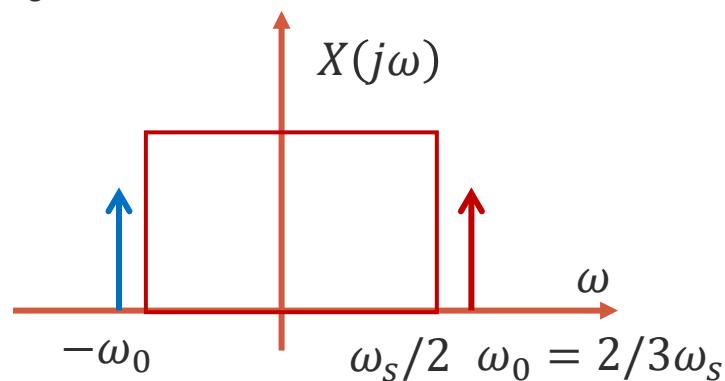
- $\omega_0 = \frac{4\omega_s}{6}$, $\omega_s = 1.5\omega_0$ (发生混叠)



信号的时域采样（欠采样）

- 设 $x(t) = \cos(\omega_0 t)$, 固定 ω_s , 考察不同 ω_0 与 ω_s 的关系时的情况

- $\omega_0 = \frac{4\omega_s}{6}, \omega_s = 1.5\omega_0$ (发生混叠)



- 原始频率 ω_0 被混叠为低频率 $\omega_s - \omega_0$

$\omega_0 < \omega_s < 2\omega_0, \omega_s \downarrow$

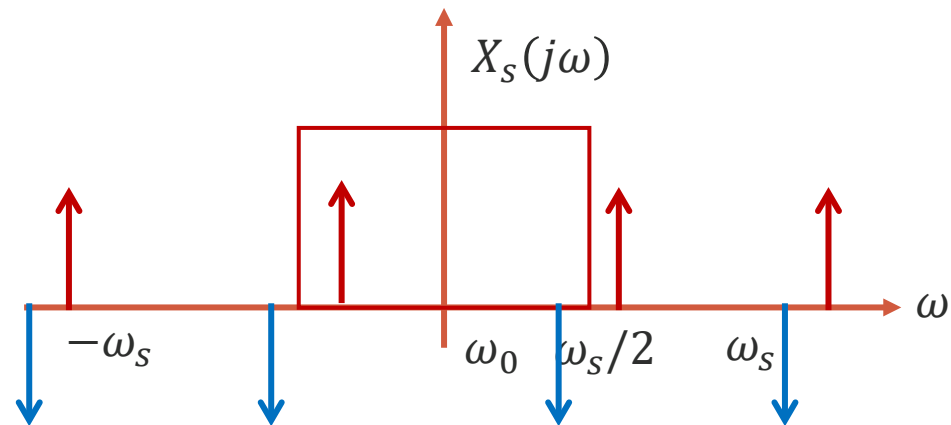
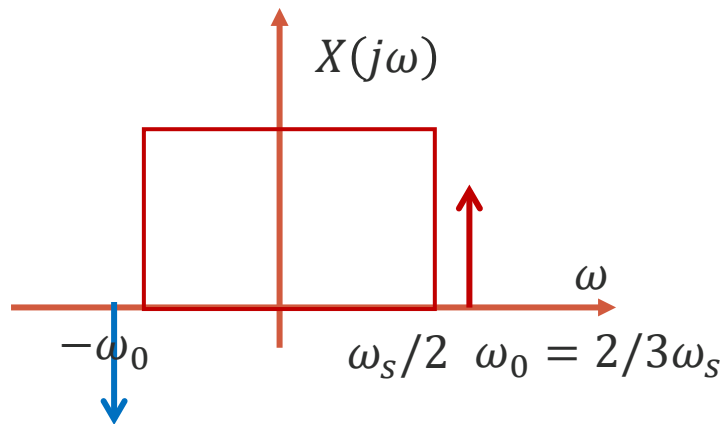
- 当 $\frac{\omega_s}{2} < \omega_0 < \omega_s$ 时, 随 ω_0 相对 ω_s 的增加, 输出频率 $\omega_s - \omega_0$ 会减小

- $\omega_s = \omega_0$ 时, 重建后的信号为常数 (每个周期只采样一次)

$f_{out} = \omega_s - \omega_0 \downarrow$

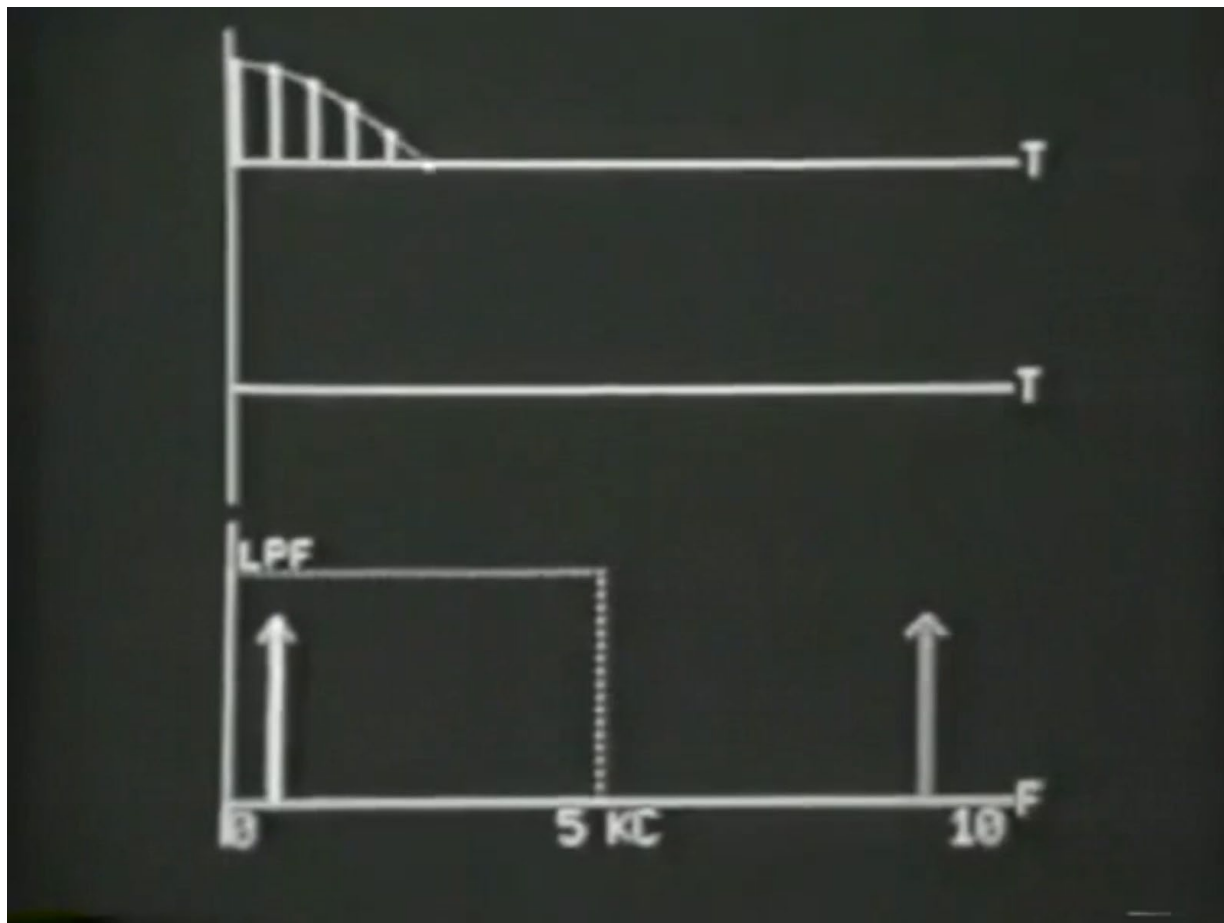
信号的时域采样（欠采样）

- 设 $x(t) = \sin(\omega_0 t)$, 固定 ω_s , 考察不同 ω_0 与 ω_s 的关系时的情况
 - $\omega_0 = \frac{4\omega_s}{6}, \omega_s = 1.5\omega_0$ (发生混叠)



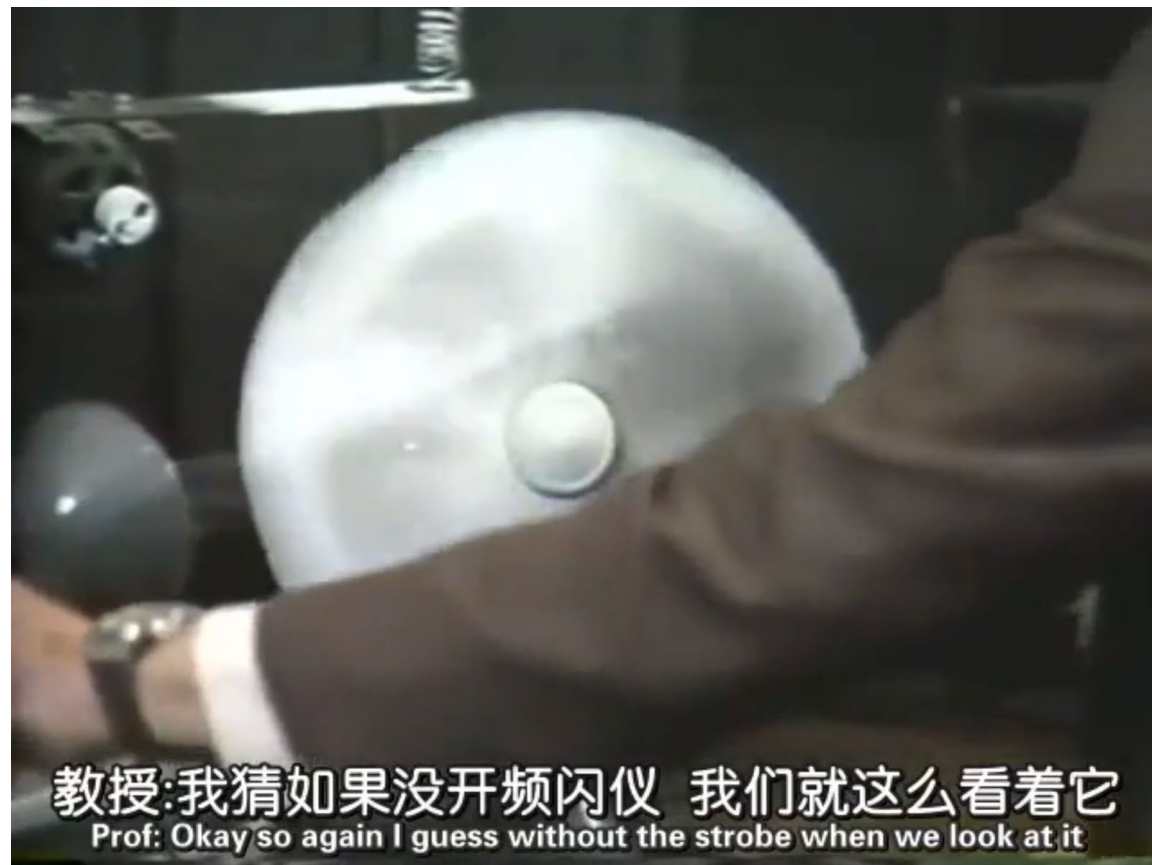
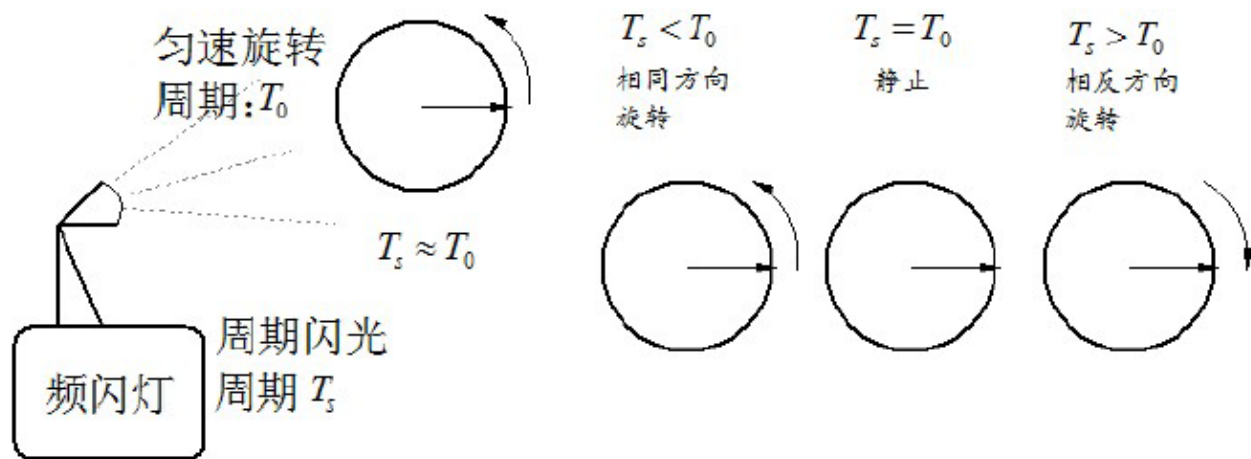
- 产生相位倒置 (phase reversal)

信号的时域采样 (欠采样)



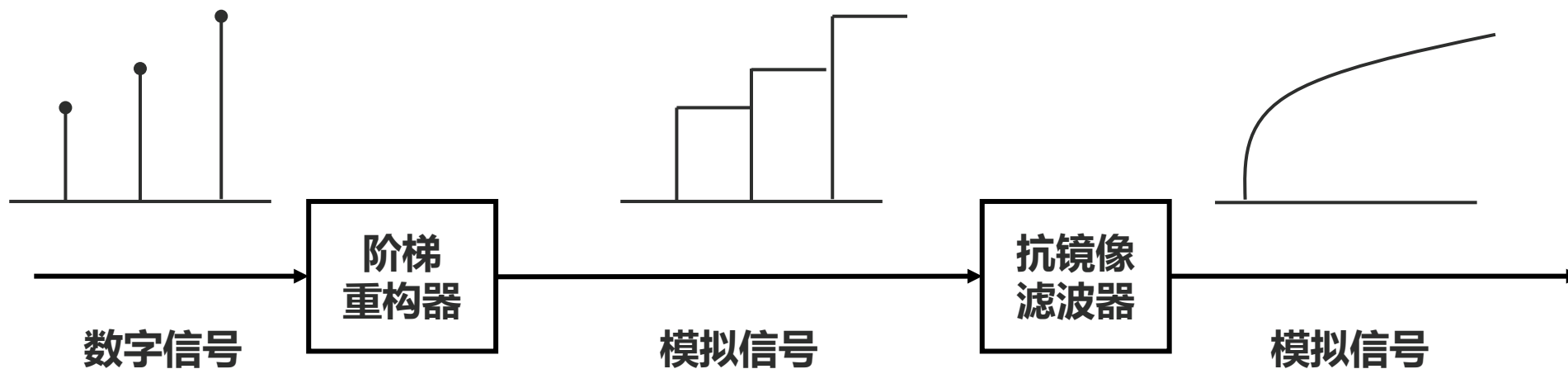
欠采样示例

- 频率混叠对低频的影响，原本是高频信号，采样后会变成低频序列，干扰原始信号的低频频谱



从数字信号到模拟信号

■ D/A的工作流程



保持信号，使当前时刻的样本值保持到下一个时刻，使信号更加光滑（滤除高频部分）

再次通过低通滤波器，进一步滤除高频分量